Skills Summary

Vertex, Axis, Intercepts and Opening Direction

Use this reference while completing your worksheet or when studying.

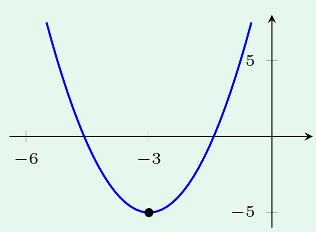
The four features, and where each one comes from:

- **Vertex** — the turning point. Vertex form $y = a(x-h)^2+k$ shows it as (h,k); standard form $y = ax^2+bx+c$ gives its x-value as $-\frac{b}{2a}$, and you substitute to get the height.
- **Axis of symmetry** — the vertical line through the vertex, $x = h$ or $x = -\frac{b}{2a}$. The graph is a mirror image across it.
- **Intercepts** — the y-intercept is c (set $x = 0$); the x-intercepts are the solutions of $y = 0$, and the axis sits midway between them.
- **Opening direction** — decided by the sign of a alone: $a > 0$ opens up (vertex is a minimum), $a < 0$ opens down (vertex is a maximum).

1. Reading features off vertex form

In $y = a(x-h)^2+k$ the vertex is (h,k) and the axis is $x = h$. The sign inside the parentheses flips: $(x+p)^2$ is $(x-(-p))^2$, so h is negative there. The size of a sets the width — bigger than one is narrower than $y = x^2$, smaller is wider.

Example: Give the vertex, the axis and the opening direction of $y = 2(x+3)^2 - 5$.
Step 1. The equation is already in vertex form, so no algebra is needed — read h and k off it, then let the sign of a settle the direction.
Step 2. $(x+3)^2 = (x-(-3))^2$, so $h = -3$ and the axis is $x = -3$; the constant outside gives k. Substituting confirms it: $y(-3) = 2(0)^2 - 5$.



Vertex height at $x = -3$: $\mathbf{-5}$, so the vertex is $(-3,-5)$ and, since a is positive, the parabola opens up.

Try it: For $y = -(x-4)^2+7$, what is the y-value at the vertex, and does the parabola open up or down?

check: **7** at $x = 4$; opens down

2. Reading features off standard form

Find the axis first: $x = -\frac{b}{2a}$. Keep *both* minus signs — the one in the formula and any minus already on b or on a. Substituting that x back into the equation gives the vertex height, and c is the y-intercept for free.

Example: Find the vertex and the axis of $y = x^2 - 10x + 21$.
Step 1. Nothing here is in vertex form and the intercepts are not obvious, so go through the axis-of-symmetry formula rather than hunting for factors.
Step 2. $a = 1$ and $b = -10$, so the axis is $x = -\dfrac{-10}{2(1)} = 5$. Substituting: $y(5) = 25 - 50 + 21$.
Vertex height at $x = 5$: $\mathbf{-4}$, so the vertex is $(5, -4)$, the axis is $x = 5$, and the y-intercept is 21.

Try it: Find the vertex of $y = x^2 + 8x + 7$. (Axis first, then substitute.)

check: axis $x = -4$, vertex height $\mathbf{-9}$

3. Using the intercepts and the sign of a

The x-intercepts are the solutions of $y = 0$, so factor when you can: from $(x-p)(x-q)$ they are p and q. The axis of symmetry sits exactly midway between them, which is a second, independent route to the vertex — and a free check on $-\frac{b}{2a}$. A parabola may have two, one or no x-intercepts, but it always has exactly one y-intercept.

Example: Find the x-intercepts of $y = x^2 + 2x - 15$ and use them to locate the axis.
Step 1. The question asks where the graph meets the x-axis, which is the same as solving $y = 0$ — so factor rather than complete the square.
Step 2. Two numbers with product -15 and sum 2 are 5 and -3: $x^2+2x-15 = (x+5)(x-3)$, so $x + 5 = 0$ or $x - 3 = 0$. x-intercepts: $\boldsymbol{x = -5}$ and $\boldsymbol{x = 3}$, whose midpoint puts the axis at $x = -1$; a is positive, so the curve opens up and that midpoint is a minimum.

Try it: Find the x-intercepts of $y = x^2 - 6x + 8$, then say where the axis of symmetry is.

check: $\boldsymbol{x = 2}$ and $\boldsymbol{x = 4}$, axis $x = 3$

(!) Watch out: $y = (x + 5)^2$ has its vertex at $x = -5$, not $x = 5$ — the sign inside the parentheses always flips. And in $-\frac{b}{2a}$ the sign of a counts too: for $y = -x^2 + 6x - 5$ the axis is $x = -\frac{6}{-2} = 3$, not -3.